Assignment 7

223010195@link.cuhk.edu.cn;221019048@link.cuhk.edu.cn

Please note that

- **Released date: November 22th, Friday.**
- **Due date: December 4th, Wednesday, by 11:55pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS7.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Find the constants a and b such that the vectors $\mathbf{u} = \begin{bmatrix} a \\ 4 \\ -b \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} a \\ 1 \\ b \end{bmatrix}$ are orthogonal and a=b+1.

Question 2. Let $x = (4, 4, -4, 4)^T$ and $y = (4, 2, 2, 1)^T$.

(a) Determine the angle between x and y.

(b) Determine the distance between x and y.

Question 3. For the following pair of vectors $\mathbf{x}$ and $\mathbf{y}$, find the vectors projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$ and verify that $\mathbf{p}$ and $\mathbf{x}$-$\mathbf{p}$ are orthogonal:

$$\mathbf{x} = (3, 5)^T, \mathbf{y} = (1, 1)^T$$

Question 4. Let $\mathbf{x}$ and $\mathbf{y}$ be linearly independent vectors in $\mathbb{R}^n$ and let S =Span$\{\mathbf{x}, \mathbf{y}\}$. Construct a matrix by $\boldsymbol{A}$= $\mathbf{x}\mathbf{y}^T + \mathbf{y}\mathbf{x}^T$.

(a) Show that $\boldsymbol{A}$ is symmetric.

(b) Show that Null($\boldsymbol{A}$)=$S^{\perp}$.

Question 5. (a) Let S be the subspace of $\mathbb{R}^3$ spanned by the vectors $x = (x_1, x_2, x_3)^T$ and $y = (y_1, y_2, y_3)^T$. Let

$$A = \begin{bmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \end{bmatrix}.$$

7_8

Show that $S^\perp = \text{Null}(A)$.

(b) Find the orthogonal complement of the subspace of $\mathbb{R}^3$ spanned by $(1, 2, 1)^T$ and $(1, -1, 2)^T$.

Question 6. Let $\mathbf{x} = (1, 1, 1, 1)^T$ and $\mathbf{y} = (8, 2, 2, 0)^T$.

7_11

(a) Determine the angle θ between $\mathbf{x}$ and $\mathbf{y}$.

(b) Find the vector projection $\mathbf{p}$ of $\mathbf{x}$ onto $\mathbf{y}$.

(c) Verify that $\mathbf{x} - \mathbf{p}$ is orthogonal to $\mathbf{p}$.

(d) Compute $\|\mathbf{x} - \mathbf{p}\|, \|\mathbf{p}\|, \|\mathbf{x}\|$ and verify that the Pythagorean law is satisfied.

Question 7. Let $\mathbf{x}$ and $\mathbf{y}$ be vectors in $\mathbb{R}^n$ and define

$$\mathbf{p} = \frac{\mathbf{x}^T\mathbf{y}}{\mathbf{y}^T\mathbf{y}}\mathbf{y} \quad \text{and} \quad \mathbf{z} = \mathbf{x} - \mathbf{p}$$

7_15

(a) Show that $\mathbf{p} \perp \mathbf{z}$. Thus, $\mathbf{p}$ is the vector projection of $\mathbf{x}$ onto $\mathbf{y}$; that is, $\mathbf{x} = \mathbf{p} + \mathbf{z}$, where $\mathbf{p}$ and $\mathbf{z}$ are orthogonal components of $\mathbf{x}$, and $\mathbf{p}$ is a scalar multiple of $\mathbf{y}$.

(b) If $\|\mathbf{p}\| = 6$ and $\|\mathbf{z}\| = 8$, determine the value of $\|\mathbf{x}\|$.

Question 8. Let S be the subspace of $\mathbb{R}^n$ spanned by the vectors $\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_k$. Show that $\mathbf{y} \in S^\perp$ if and only if $\mathbf{y} \perp \mathbf{x}_i$ for $i = 1, \ldots, k$.

7_16

Question 9. For each of the following systems $A\mathbf{x} = \mathbf{b}$, find all least squares solutions:

7_9

(a) $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \\ -1 & -2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$

(b) $A = \begin{pmatrix} 1 & 1 & 3 \\ -1 & 3 & 1 \\ 1 & 2 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} -2 \\ 0 \\ 8 \end{pmatrix}$

Question 10. Which of the following sets of vectors form an orthonormal basis for $\mathbb{R}^2$?

8_1

(a) $\left\{ \left(\frac{3}{5}, \frac{4}{5}\right)^T, \left(\frac{5}{13}, \frac{12}{13}\right)^T \right\}$

(b) $\{(1,-1)^T,(1,1)^T\}$

(c) $\left\{\left(\frac{\sqrt{3}}{2},\frac{1}{2}\right)^T,\left(-\frac{1}{2},\frac{\sqrt{3}}{2}\right)^T\right\}$

Question 11. Let

$$\mathbf{u}_1=\begin{pmatrix}\frac{1}{3\sqrt{2}}\\ \frac{1}{3\sqrt{2}}\\ -\frac{4}{3\sqrt{2}}\end{pmatrix},\quad \mathbf{u}_2=\begin{pmatrix}\frac{2}{3}\\ \frac{2}{3}\\ \frac{1}{3}\end{pmatrix},\quad \mathbf{u}_3=\begin{pmatrix}\frac{1}{\sqrt{2}}\\ -\frac{1}{\sqrt{2}}\\ 0\end{pmatrix}$$

(a) Show that $\{\mathbf{u}_1,\mathbf{u}_2,\mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x}=(1,1,1)^T$. Write $\mathbf{x}$ as a linear combination of $\mathbf{u}_1,\mathbf{u}_2$, and $\mathbf{u}_3$, and compute $\|\mathbf{x}\|$.

Question 12. Let $\{\mathbf{u}_1,\mathbf{u}_2,\mathbf{u}_3\}$ be an orthonormal basis for an inner product space V. If $\mathbf{x}=c_1\mathbf{u}_1+c_2\mathbf{u}_2+c_3\mathbf{u}_3$ is a vector with the properties $\|\mathbf{x}\|=5,\langle\mathbf{u}_1,\mathbf{x}\rangle=4$, and $\mathbf{x}\perp\mathbf{u}_2$, then what are the possible values of c_1,c_2,c_3 ?

Question 13. For each of the following, use the Gram-Schmidt process to find an orthonormal basis for Row(A).

(a) $A=\begin{pmatrix}-1 & 3\\ 1 & 5\end{pmatrix}$

(b) $A=\begin{pmatrix}2 & 5\\ 1 & 10\end{pmatrix}$

Question 14. Let $A=\begin{pmatrix}2 & 1\\ 1 & 1\\ 2 & 1\end{pmatrix}$ and $\mathbf{b}=\begin{pmatrix}12\\ 6\\ 18\end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of A.

(b) Factor A into a product QR, where Q has an orthonormal set of column vectors and R is upper triangular.

(c) Solve the least squares problem $A\mathbf{x}=\mathbf{b}$.

Question 15. Let V be an inner product space over $\mathbb{R}$, $\forall u,v\in V$, show the following:

(a) $\langle u,v\rangle=\frac{1}{4}\|u+v\|^2-\frac{1}{4}\|u-v\|^2$

(b) $\|u+v\|^2+\|u-v\|^2=2\|u\|^2+2\|v\|^2$

Question 16. Given the vector space $C[-1,1]$ with inner product

$$\langle f, g \rangle = \int_{-1}^{1} f(x)g(x)dx$$

and norm

$$\|f\| = (\langle f, f \rangle)^{1/2}$$

(a) Show that the vectors 1 and x are orthogonal.

(b) Compute $\|1\|$ and $\|x\|$.